Cambridge Assessment International Education

Cambridge International AS & A Level

BIOLOGY

9700/01

Paper 2 AS Level Structured Questions – The Microscope in Cell Studies

MARK SCHEME

Maximum Mark: 134

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of 12 printed pages.

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[Turn over

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:
• the specific content of the mark scheme or the generic level descriptors for the question
• the specific skills defined in the mark scheme or in the generic level descriptors for the question
• the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always whole marks (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded positively:
• marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
• marks are awarded when candidates clearly demonstrate what they know and can do
• marks are not deducted for errors or omissions
• answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question.

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles & Abbreviations

1 Context and scientific use of keywords: marks should not be awarded if keywords are used incorrectly.
2 Contradictory statements: credit should not be awarded for any correct statement that is contradicted within the same question part.
3 Calculation guidance: correct answers to calculations should be given full credit even if there is no working, unless stated otherwise.
Abbreviations: ; separates marking points | **/** alternative answers | **R** reject | **A** accept | **AW** alternative wording | **underline** essential word | **max** maximum mark | **ecf** error carried forward | **I** ignore | **AVP** alternative valid point.

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Question	Answer	Marks
1(a)	• number of times larger an image is compared to the actual size (of the specimen) / ratio of image size to actual size / image size / actual size ; ***A*** *M = I / A* ***R*** *actual size / image size*	1
1(b)	• the ability to distinguish between two separate points (as separate entities) / the minimum distance between two points that can still be seen as distinct ; ***A*** *clarity of fine detail* ***I*** *references to magnification alone*	1
1(c)	• formula: magnification = image size / actual size / *M = I / A* (or *A = I / M*) ; • correct conversion of units (e.g. mm × 1 000 = µm / cm × 10 000 = µm) ; • correct substitution of values from diagram / text into formula ; • correct final answer with appropriate units (µm / nm) or '×' for magnification ; ***A*** *correct numerical answer alone scores full marks* ***ecf*** *error carried forward if formula is correct but conversion error occurred*	3
1(d)	• use a sharp (HB) pencil / single continuous clear lines / no sketching / no feathery lines ; • no shading / no hatching / no stippling ; • large drawing occupying at least 50% / half of the available space ; • straight / ruled label lines touching structures without arrowheads ; • correct proportions and relative positions of structures accurately shown ; ***max*** *[marks]* ***R*** *use of ink, colored pencils, or unruled label lines*	3
2(a)	• place thin layer / single-cell thick section of tissue onto slide ; • add a drop of stain (iodine solution / methylene blue) / water ; • lower coverslip at an angle using a mounted needle / forceps ; • slowly / gently to prevent trapping air bubbles ; • blot excess liquid / stain with filter paper ; ***A*** *prevents damage to objective lens*	2
2(b)	• detailed account of sequence of events / structural features observed ; • specific factual details / trends / data points cited ; ***A*** *descriptive points in any logical order*	1

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Question	Answer	Marks
2(c)	• formula: magnification = image size / actual size / $M = I / A$ (or $A = I / M$) ; • correct conversion of units (e.g. mm × 1 000 = µm / cm × 10 000 = µm) ; • correct final answer with appropriate units (µm / nm) or '×' for magnification ; ***A*** *correct numerical answer alone scores full marks* ***ecf*** *error carried forward if formula is correct but conversion error occurred*	2
2(d)	• electron microscope has higher resolution due to shorter wavelength of electrons (0.5 nm vs 200 nm for light) ; • higher useful magnification possible (> ×500 000 vs ×1 500 for light) ; • allows internal ultrastructure of organelles to be resolved (e.g. cristae, ribosomes, thylakoids) ; • disadvantage: specimen must be dead / placed in a vacuum / risk of preparation artefacts ; ***A*** *light microscope allows viewing of live specimens and true color* ***R*** *3D image unless SEM is specified*	2
3(a)	• electron microscope has higher resolution due to shorter wavelength of electrons (0.5 nm vs 200 nm for light) ; • higher useful magnification possible (> ×500 000 vs ×1 500 for light) ; • allows internal ultrastructure of organelles to be resolved (e.g. cristae, ribosomes, thylakoids) ; • disadvantage: specimen must be dead / placed in a vacuum / risk of preparation artefacts ; • further elaboration of biological principle / detail of mechanism ; ***A*** *light microscope allows viewing of live specimens and true color* ***R*** *3D image unless SEM is specified*	5
3(b)	• eyepiece graticule has an arbitrary scale / no absolute units ; • line up eyepiece graticule scale with stage micrometer scale (which has known divisions, e.g. 0.01 mm) ; • count number of graticule units corresponding to a known distance on the stage micrometer ; • calculate distance per graticule unit = stage micrometer distance / number of graticule units ; • calibration must be repeated at each magnification / with each objective lens ; ***A*** *1 epu = distance / units*	2
3(c)	• correct biological term / structure identified ; • further elaboration of biological principle / detail of mechanism ; ***R*** *ambiguous terminology*	2
3(d)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	1

Question	Answer	Marks
4(a)	• use a sharp (HB) pencil / single continuous clear lines / no sketching / no feathery lines ; • no shading / no hatching / no stippling ; • large drawing occupying at least 50% / half of the available space ; • straight / ruled label lines touching structures without arrowheads ; • correct proportions and relative positions of structures accurately shown ; ***max*** *[marks]* ***R*** *use of ink, colored pencils, or unruled label lines*	**5**
4(b)	• use a sharp (HB) pencil / single continuous clear lines / no sketching / no feathery lines ; • no shading / no hatching / no stippling ; • large drawing occupying at least 50% / half of the available space ; • straight / ruled label lines touching structures without arrowheads ; • correct proportions and relative positions of structures accurately shown ; ***max*** *[marks]* ***R*** *use of ink, colored pencils, or unruled label lines*	**5**
4(c)	• valid scientific hypothesis / plausible biological explanation ; • application of syllabus knowledge to novel scenario ; ***A*** *reasonable alternative suggestions supported by biological principles*	**1**
4(d)	• correct biological term / structure identified ; ***R*** *ambiguous terminology*	**1**
5(a)	• formula: magnification = image size / actual size / $M = I / A$ (or $A = I / M$) ; • correct conversion of units (e.g. mm × 1 000 = µm / cm × 10 000 = µm) ; • correct final answer with appropriate units (µm / nm) or '×' for magnification ; ***A*** *correct numerical answer alone scores full marks* ***ecf*** *error carried forward if formula is correct but conversion error occurred*	**1**
5(b)	• valid scientific hypothesis / plausible biological explanation ; • application of syllabus knowledge to novel scenario ; ***A*** *reasonable alternative suggestions supported by biological principles*	**1**

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Question	Answer	Marks
5(c)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	**2**
5(d)	• electron microscope has higher resolution due to shorter wavelength of electrons (0.5 nm vs 200 nm for light) ; • higher useful magnification possible (> ×500 000 vs ×1 500 for light) ; • allows internal ultrastructure of organelles to be resolved (e.g. cristae, ribosomes, thylakoids) ; • disadvantage: specimen must be dead / placed in a vacuum / risk of preparation artefacts ; ***A*** *light microscope allows viewing of live specimens and true color* ***R*** *3D image unless SEM is specified*	**1**
1(a)	• formula: magnification = image size / actual size / $M = I / A$ (or $A = I / M$) ; • correct conversion of units (e.g. mm × 1 000 = µm / cm × 10 000 = µm) ; • correct final answer with appropriate units (µm / nm) or '×' for magnification ; ***A*** *correct numerical answer alone scores full marks* ***ecf*** *error carried forward if formula is correct but conversion error occurred*	**2**
1(b)	• formula: magnification = image size / actual size / $M = I / A$ (or $A = I / M$) ; • correct conversion of units (e.g. mm × 1 000 = µm / cm × 10 000 = µm) ; • correct final answer with appropriate units (µm / nm) or '×' for magnification ; ***A*** *correct numerical answer alone scores full marks* ***ecf*** *error carried forward if formula is correct but conversion error occurred*	**2**
1(c)	• formula: magnification = image size / actual size / $M = I / A$ (or $A = I / M$) ; • correct conversion of units (e.g. mm × 1 000 = µm / cm × 10 000 = µm) ; • correct final answer with appropriate units (µm / nm) or '×' for magnification ; ***A*** *correct numerical answer alone scores full marks* ***ecf*** *error carried forward if formula is correct but conversion error occurred*	**2**
1(d)	• accurate biological statement meeting the syllabus learning objective ; • supporting detail / elaboration of biological concept ; ***A*** *acceptable syllabus alternative [AW]*	**2**

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Question	Answer	Marks
2(a)	• detailed account of sequence of events / structural features observed ; • specific factual details / trends / data points cited ; • comprehensive description addressing all aspects of the process / observation ; ***A*** *descriptive points in any logical order*	3
2(b)	• detailed account of sequence of events / structural features observed ; • specific factual details / trends / data points cited ; • comprehensive description addressing all aspects of the process / observation ; • further elaboration of biological principle / detail of mechanism ; ***A*** *descriptive points in any logical order*	4
2(c)	• correct biological term / structure identified ; • further elaboration of biological principle / detail of mechanism ; ***R*** *ambiguous terminology*	2
3(a)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	2
3(b)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	1
3(c)	• accurate biological statement meeting the syllabus learning objective ; • supporting detail / elaboration of biological concept ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; ***A*** *acceptable syllabus alternative [AW]*	4

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Question	Answer	Marks
3(d)	• electron microscope has higher resolution due to shorter wavelength of electrons (0.5 nm vs 200 nm for light) ; • higher useful magnification possible (> ×500 000 vs ×1 500 for light) ; • allows internal ultrastructure of organelles to be resolved (e.g. cristae, ribosomes, thylakoids) ; • disadvantage: specimen must be dead / placed in a vacuum / risk of preparation artefacts ; ***A*** *light microscope allows viewing of live specimens and true color* ***R*** *3D image unless SEM is specified*	**2**
4(a)	• use a sharp (HB) pencil / single continuous clear lines / no sketching / no feathery lines ; • no shading / no hatching / no stippling ; • large drawing occupying at least 50% / half of the available space ; • straight / ruled label lines touching structures without arrowheads ; • correct proportions and relative positions of structures accurately shown ; ***max*** *[marks]* ***R*** *use of ink, colored pencils, or unruled label lines*	**4**
4(b)	• formula: magnification = image size / actual size / $M = I / A$ (or $A = I / M$) ; • correct conversion of units (e.g. mm × 1 000 = µm / cm × 10 000 = µm) ; • correct final answer with appropriate units (µm / nm) or '×' for magnification ; ***A*** *correct numerical answer alone scores full marks* ***ecf*** *error carried forward if formula is correct but conversion error occurred*	**2**
4(c)	• electron microscope has higher resolution due to shorter wavelength of electrons (0.5 nm vs 200 nm for light) ; • higher useful magnification possible (> ×500 000 vs ×1 500 for light) ; • allows internal ultrastructure of organelles to be resolved (e.g. cristae, ribosomes, thylakoids) ; • disadvantage: specimen must be dead / placed in a vacuum / risk of preparation artefacts ; ***A*** *light microscope allows viewing of live specimens and true color* ***R*** *3D image unless SEM is specified*	**2**
5(a)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	**2**

Question	Answer	Marks
5(b)	• use a sharp (HB) pencil / single continuous clear lines / no sketching / no feathery lines ; • no shading / no hatching / no stippling ; • large drawing occupying at least 50% / half of the available space ; • straight / ruled label lines touching structures without arrowheads ; • correct proportions and relative positions of structures accurately shown ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; ***max*** *[marks]* ***R*** *use of ink, colored pencils, or unruled label lines*	**8**
5(c)	• formula: magnification = image size / actual size / $M = I / A$ (or $A = I / M$) ; • correct conversion of units (e.g. mm × 1 000 = µm / cm × 10 000 = µm) ; • correct final answer with appropriate units (µm / nm) or '×' for magnification ; ***A*** *correct numerical answer alone scores full marks* ***ecf*** *error carried forward if formula is correct but conversion error occurred*	**2**
5(d)	• place thin layer / single-cell thick section of tissue onto slide ; • add a drop of stain (iodine solution / methylene blue) / water ; • lower coverslip at an angle using a mounted needle / forceps ; • slowly / gently to prevent trapping air bubbles ; • blot excess liquid / stain with filter paper ; ***A*** *prevents damage to objective lens*	**3**
1(a)	• eyepiece graticule has an arbitrary scale / no absolute units ; • line up eyepiece graticule scale with stage micrometer scale (which has known divisions, e.g. 0.01 mm) ; • count number of graticule units corresponding to a known distance on the stage micrometer ; • calculate distance per graticule unit = stage micrometer distance / number of graticule units ; • calibration must be repeated at each magnification / with each objective lens ; ***A*** *1 epu = distance / units*	**2**
1(b)	• eyepiece graticule has an arbitrary scale / no absolute units ; • line up eyepiece graticule scale with stage micrometer scale (which has known divisions, e.g. 0.01 mm) ; • count number of graticule units corresponding to a known distance on the stage micrometer ; • calculate distance per graticule unit = stage micrometer distance / number of graticule units ; • calibration must be repeated at each magnification / with each objective lens ; ***A*** *1 epu = distance / units*	**3**

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Question	Answer	Marks
1(c)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	2
1(d)	• eyepiece graticule has an arbitrary scale / no absolute units ; • line up eyepiece graticule scale with stage micrometer scale (which has known divisions, e.g. 0.01 mm) ; • count number of graticule units corresponding to a known distance on the stage micrometer ; • calculate distance per graticule unit = stage micrometer distance / number of graticule units ; • calibration must be repeated at each magnification / with each objective lens ; ***A*** *1 epu = distance / units*	1
2(a)	• accurate biological statement meeting the syllabus learning objective ; • supporting detail / elaboration of biological concept ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; ***A*** *acceptable syllabus alternative [AW]*	4
2(b)	• correct biological term / structure identified ; ***R*** *ambiguous terminology*	1
2(c)	• formula: magnification = image size / actual size / $M = I / A$ (or $A = I / M$) ; • correct conversion of units (e.g. mm × 1 000 = µm / cm × 10 000 = µm) ; • correct final answer with appropriate units (µm / nm) or '×' for magnification ; ***A*** *correct numerical answer alone scores full marks* ***ecf*** *error carried forward if formula is correct but conversion error occurred*	2
2(d)	• valid scientific hypothesis / plausible biological explanation ; • application of syllabus knowledge to novel scenario ; ***A*** *reasonable alternative suggestions supported by biological principles*	2

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Question	Answer	Marks
3(a)	• place thin layer / single-cell thick section of tissue onto slide ; • add a drop of stain (iodine solution / methylene blue) / water ; • lower coverslip at an angle using a mounted needle / forceps ; • slowly / gently to prevent trapping air bubbles ; • blot excess liquid / stain with filter paper ; ***A*** *prevents damage to objective lens*	**3**
3(b)	• accurate biological statement meeting the syllabus learning objective ; ***A*** *acceptable syllabus alternative [AW]*	**1**
3(c)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; • detailed biochemical context: reference to specific molecular interactions / syllabus concepts ; • further elaboration of biological principle / detail of mechanism ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	**4**
3(d)	• eyepiece graticule has an arbitrary scale / no absolute units ; • line up eyepiece graticule scale with stage micrometer scale (which has known divisions, e.g. 0.01 mm) ; • count number of graticule units corresponding to a known distance on the stage micrometer ; • calculate distance per graticule unit = stage micrometer distance / number of graticule units ; • calibration must be repeated at each magnification / with each objective lens ; ***A*** *1 epu = distance / units*	**3**
4(a)	• formula: magnification = image size / actual size / $M = I / A$ (or $A = I / M$) ; • correct conversion of units (e.g. mm × 1 000 = µm / cm × 10 000 = µm) ; • correct final answer with appropriate units (µm / nm) or '×' for magnification ; ***A*** *correct numerical answer alone scores full marks* ***ecf*** *error carried forward if formula is correct but conversion error occurred*	**2**
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Question	Answer	Marks
4(c)	• use a sharp (HB) pencil / single continuous clear lines / no sketching / no feathery lines ; • no shading / no hatching / no stippling ; • large drawing occupying at least 50% / half of the available space ; • straight / ruled label lines touching structures without arrowheads ; • correct proportions and relative positions of structures accurately shown ; ***max*** *[marks]* ***R*** *use of ink, colored pencils, or unruled label lines*	**4**
5(a)	• formula: magnification = image size / actual size / $M = I / A$ (or $A = I / M$) ; • correct conversion of units (e.g. mm × 1 000 = µm / cm × 10 000 = µm) ; • correct substitution of values from diagram / text into formula ; • correct final answer with appropriate units (µm / nm) or '×' for magnification ; ***A*** *correct numerical answer alone scores full marks* ***ecf*** *error carried forward if formula is correct but conversion error occurred*	**3**
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5(c)	• correct biological term / structure identified ; • further elaboration of biological principle / detail of mechanism ; ***R*** *ambiguous terminology*	**2**

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